

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CASE STUDY

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	20% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	AKSHAYA KUMAR CHANDRASEKAR	Reg. No.:	192425350
Section / Batch:	B.TECH AIML	Date:	06/08/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues, provide an in-depth analysis, and propose innovative, feasible solutions with strong justification.

Question Type	Focus Area	Questions
Case Study	Focus on Design Divide and Conquer algorithms: Merge Sort, Quick Sort, Binary Search and Matrix Multiplication	Q1

SECTION A – CASE STUDY

```
1 import random
2
3 def quicksort_3way(a):
4     if len(a) <= 1:
5         return a
6     stack = [(0, len(a) - 1)]
7
8     while stack:
9         lo, hi = stack.pop()
10        if lo >= hi:
11            continue
12
13        pivot = a[random.randint(lo, hi)]
14        lt = lo
15        i = lo
16        gt = hi
17
18        while i <= gt:
19            if a[i] < pivot:
20                a[lt], a[i] = a[i], a[lt]
21                lt += 1
22                i += 1
23            elif a[i] > pivot:
24                a[i], a[gt] = a[gt], a[i]
25                gt -= 1
26            else:
27                i += 1
28        left_size = (lt - 1) - lo
29        right_size = hi - (gt + 1)
```

stdin input

Output

```
Before: [120, 99.99, 110, 99.99,
130, 105, 99.99, 115]
After:  [99.99, 99.99, 99.99, 105,
110, 115, 120, 130]
```

Code of Conduct

I AKSHAYA KUMAR CHANDRASEKAR(192425350)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: AKSHAYA KUMAR.C

RUBRICS FOR CALCULATION

Case Study					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25%	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts

Analysis	20%	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20%	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10%	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%				
Application of Concepts	25%				
Analysis	20%				
Solution Design	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____